

Anna O'Grady

NASA HUBBLE POSTDOCTORAL FELLOW – CARNEGIE MELLON UNIVERSITY & THE UNIVERSITY OF PITTSBURGH

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Research Skills and Interests

Topics: Observational Stellar Astronomy, Binary Massive Star Evolution, Resolved Stellar Populations, Variable Stars, Core Collapse Supernovae Progenitors, UVOIR Spectroscopy, Population Synthesis Modeling

Telescopes: Hubble Space Telescope, Chandra X-ray Observatory, Magellan Telescopes (LCO), Gemini, South African Large Telescope

Software: COSMIC, DRAGONS, IRAF, AstroPy, Python, Git

Research Positions

Postdoctoral Fellow

NASA HUBBLE FELLOWSHIP PROGRAM

Carnegie Mellon University &
The University of Pittsburgh
Sept. 2026 - Present

Postdoctoral Fellow

McWILLIAMS CENTER POSTDOCTORAL FELLOWSHIP

Carnegie Mellon University
Sept. 2023 - Aug. 2026

PhD Student

ADVISORS: PROF. MARIA DROUT, PROF. BRYAN GAENSLER, AND PROF RENÉE HLÔZEK

University of Toronto
Sept. 2016 - Aug. 2023

Undergraduate Thesis, MUN & NSERC Research Award

ADVISORS: PROF. IVAN BOOTH AND PROF. HARI KUNDURI

Memorial University of NL
May. 2015 - August 2016

Dorrit Hoffleit Undergraduate Summer Research Award

ADVISOR: PROF. LOUISE EDWARDS

Yale University
May 2014 - July 2014

Education

University of Toronto

PHD IN ASTRONOMY & ASTROPHYSICS

Advisors: Prof. Maria Drout and Prof. Bryan Gaensler

Title: Identification of super-AGB stars and Yellow Supergiant Binaries in the Magellanic Clouds

Toronto, ON, Canada
2016-2023

Memorial University of Newfoundland and Labrador

BSc HONOURS IN PHYSICS & APPLIED MATHEMATICS

Advisors: Prof. Ivan Booth and Prof. Hari Kunduri

Title: Perturbations of black hole horizons

St. John's, NL, Canada
2012-2016

Advising Experience

3 students supervised. (°) indicates co-supervision.

PhD	Shannon Bowes °, PhD Thesis: Variability of Yellow Supergiant Binaries	2025-Present
BSc	Mark Martinez °, Project: SNe Progenitors of Binary Neutron Star Mergers in COSMIC	2024-Present
High Sch.	Esme Offner , Science Fair: Radial Velocities of Nearby Stars	2024-2025

Awards & Grants

SCHOLARSHIPS & FELLOWSHIPS

PhD	GM Women in Science and Mathematics Award, \$5000 CAD	2022
PhD	Lachlan Gilchrist Fellowship, \$12000 over 3 years	2018-2021
PhD	Walter C. Sumner Memorial Fellowship, \$16000 CAD over 2 years	2018-2020
PhD	OGS-QEII Graduate Scholarship, \$60000 CAD over 4 years	2016-2020
Undergrad	NSERC USRA & MUNL Internship Award, \$12000 CAD for 2 summers	2015-2016
Undergrad	Dr. S.W. Brekon Scholarship in Physics, \$1000 CAD	2015
Undergrad	Dorrit Hoffleit Summer Undergraduate Research Award, \$3000 USD	2014

ACADEMIC, RESEARCH, & LEADERSHIP AWARDS

PhD	Jiu Lin Allen Yen Award for O'Grady et al. 2020, \$1000 CAD	2021
PhD	"You Lead, We Follow" Leadership Award, UofT Dept. of Astronomy and Astrophysics	2021
Undergrad	Memorial University Lou Visentin Award, Achieved Dean's List in all years	2016
Undergrad	Dr. Hugh J. Anderson Leadership Award, Memorial University of NL	2013-2015

GRANTS

Postdoc	NSF Proposal - Near Field Population Synthesis, Collaborator	2025
Postdoc	McWilliams Seed Grant - Write Together Program, Yearly funding for new dept. professional development initiative	2023-2025

Observing Programs

SUCCESSFUL OBSERVING PROPOSALS

[PI] Characterizing the hot companions of the first identified Yellow supergiant binary population	Hubble Space Telescope
Co-IS: DROUT, M., BREIVIK, K.; GOTBERG, Y.; LUDWIG, B.; BOWES, S. (AWARDED: 6 ORBITS)	2026, Cycle 34
[Co-I] Catch Them While They're Hot (And Then Not): An HST Search for Hot Failed Supernovae	Hubble Space Telescope
PI: DROUT, M. (AWARDED: 40 ORBITS)	2026, Cycle 34
[Co-I] Completing a Legacy Dataset with Deep HST Imaging of 121 Nearby Star-Forming Galaxies	Hubble Space Telescope
PIs: KILPATRICK, C.; DROUT, M., FOX, O. (AWARDED: 193 ORBITS)	2026, Cycle 34
[Co-PI] Searching for super-AGB stars in the Magellanic Clouds	SOAR
Co-PIs: ROSE, S (AWARDED: 1 NIGHT)	2026B
[PI] A Complete Sample of YSG Binaries in the Magellanic Clouds: Towards a Population of Partially Stripped Supernovae Progenitors	Gemini-S
Co-IS: DROUT, M., BREIVIK, K.; GOTBERG, Y.; LUDWIG, B.; BOWES, S. (AWARDED: 17.8 HOURS)	2026B
[PI, Director's Discretionary Time] X-ray detection of the low-mass companion responsible for Betelgeuse's long secondary period	Chandra X-Ray Observatory
Co-IS: O'CONNOR, B; GOLDBERG, J.; JOYCE, M.; MOLNAR, L.; JOHNSON, C.; HARE, J.; BREIVIK, K.; DROUT, M. (AWARDED: 45 KS)	2024B, Cycle 26 DDT
[Co-I, Director's Discretionary Time] Far-UV detection of the low-mass companion responsible for Betelgeuse's long secondary period	Hubble Space Telescope
PI: JOYCE, M. (AWARDED: 4 ORBITS)	2024B, Cycle 31 DDT

[PI] Confirming YSG Binaries in the Magellanic Clouds: Towards a Population of Partially Stripped Supernovae Progenitors	<i>SALT</i>
Co-IS: DROUT, M. (AWARDED: 6.3 HOURS)	<i>2024B</i>
[PI] Confirming YSG Binaries in the Magellanic Clouds: Towards a Population of Partially Stripped Supernovae Progenitors	<i>Gemini-S</i>
Co-IS: DROUT, M.; BREIVIK, K.; GOTBERG, Y.; LUDWIG, B., (AWARDED: 22 HOURS)	<i>2024B</i>
[PI] Confirmation of YSG Binaries in the Magellanic Clouds: Towards a Population of Supernova Progenitor Analogs	<i>Gemini-S</i>
Co-IS: DROUT, M.; BREIVIK, K.; GOTBERG, Y.; LUDWIG, B., (AWARDED: 18 HOURS)	<i>2022B</i>
[Co-I] The Nature of a New Class of Luminous Highly Variable Stars	<i>Magellan/Baade 6.5m</i>
PI: DROUT, M. (AWARDED: 1 NIGHT)	<i>2020A</i>
[Co-I] An APOGEE-2S Survey of Evolved Massive Stars in the Magellanic Clouds	<i>du Pont 2.5m</i>
PI: DROUT, M. (AWARDED: 4 NIGHTS)	<i>2019</i>
[Co-I] Yellow Supergiants as Probes of Stellar Evolution and Mass Loss	<i>Magellan/Clay 6.5m</i>
PI: DROUT, M. (AWARDED: 12 NIGHTS)	<i>2017B</i>

OBSERVING EXPERIENCE

Optical Echelle Spectrographs (MIKE, MagE)	<i>Magellan 6.5m, LCO</i>
25 NIGHTS	
Optical Long-Slit Spectroscopy and Photometry (IMACS, LDSS)	<i>Magellan 6.5m, LCO</i>
4 NIGHTS	
Optical Multi-Object Spectroscopy and Photometry (WFCCD)	<i>du Pont 2.5m, LCO</i>
3 NIGHTS	

Academic & Public Presentations

COLLOQUIA & SEMINARS (13)

State Coll., US	Penn State University, Seminar	<i>Oct. 2026</i>
Keele, UK	Keele University, Seminar	<i>May. 2026</i>
Pittsburgh, US	University of Pittsburgh, Seminar	<i>Feb. 2026</i>
Hamilton, CA	McMaster University, Seminar	<i>Oct. 2025</i>
St. John's, CA	Memorial University of Newfoundland and Labrador, Colloquium	<i>Oct. 2025</i>
London, CA	University of Western Ontario, Colloquium	<i>Sep. 2025</i>
Montreal, CA	McGill University, Trottier Space Institute, TSI Seminar Series	<i>Sep. 2025</i>
St. John's, CA	Memorial University of Newfoundland and Labrador, Colloquium	<i>Mar. 2024</i>
Cambridge, US	Harvard Center for Astrophysics, Seminar	<i>Nov. 2022</i>
Toronto, CA	University of Toronto, Seminar	<i>Oct. 2022</i>
Virtual	California Polytechnic State University, Colloquium	<i>Apr. 2021</i>
Virtual	Univ. of Arizona, Univ. of Hawaii, Ohio State Univ., Paper discussion (3)	<i>Oct. 2020</i>
St. John's, CA	Memorial University of Newfoundland and Labrador, Colloquium	<i>Apr. 2019</i>

CONFERENCES & WORKSHOPS (3 INVITED TALKS, 8 CONTRIBUTED TALKS, 6 POSTERS)

Phoenix, US	AAS 247 , Contributed Talk, AAS Winter Meeting	Jan. 2026
Halifax, CA	CASCA AGM , Contributed Talk, St. Mary's University	Jun. 2025
NYC, US	Stable Mass Transfer 2.0 , <i>Invited</i> Workshop, Center for Computational Astrophysics	May 2025
Virtual	XMC II: Clouds over Yellowstone , Contributed Poster, NOIRLab	May 2025
Virtual	Transients From Space , Contributed Poster, Space Telescope Science Institute	Apr. 2025
Virtual	Stripped Star Workshop , Contributed Talk, Lorentz Center	July 2024
Sacramento, US	APS April Meeting , <i>Invited</i> Talk, Special Session on Thorne-Zytkow Objects	Apr. 2024
Leuven, BE	Massive Triples, Binaries, and Mergers , Contributed Talk, KU Leuven	July 2023
Seattle, US	AAS 241 , Dissertation Talk, AAS Winter Meeting	Jan. 2023
Toronto, CA	Spoken-WERRD symposium , Contributed Talk	Nov. 2022
Ballyconnel, IR	Massive Stars Near and Far , Contributed Poster, IAU Symposium 361	May 2022
Virtual	AAS 238 , <i>Invited</i> Talk, Special Session on Thorne-Zytkow Objects	June 2021
Virtual	Unsolved Problems in Giants & Red Supergiants , Contributed Talk, IAU Symposium, LINK	June 2021
Virtual	CASCA AGM , Contributed Talk	June 2021
Virtual	CASCA AGM , Contributed Poster	May 2020
Toronto, CA	RASC and AAVSO Joint AGM , Contributed Poster, York University	June 2019
Baltimore, US	Deaths and Afterlives of Stars , Contributed Poster, Space Telescope Science Institute	Apr. 2019

PUBLIC TALKS, PANELS, & EDUCATION WORK (23)

Pittsburgh, US	Public Science/Planetarium Talk , Pittsburgh Science Center	Aug. 26
Virtual	Public Science Talk , Pittsburgh Amateur Astronomy Association	Apr. 26
Pittsburgh, US	Teen Science Cafe Speaker , Allegheny Observatory	Jan. 25
St. John's, CA	Undergraduate Career Talk , Memorial University of Newfoundland	Oct. 2025
London, CA	Undergraduate Career Talk , University of Western Ontario	Sept. 2025
Pittsburgh, US	Public Science Talk (2) , Allegheny Observatory	Jul. 24, Apr. 25
St. John's, CA	Public Science Talk (9) , Royal Astronomical Society of Canada, St. John's Chapter	Dec. 2017-25
St. John's, CA	Elizabeth R. Laird Lecture , Johnson GeoCentre	Apr. 2024
Mt. Pearl, CA	Career Talk , Mount Pearl Girl Guides Association	Apr. 2024
Virtual	Public Science Talk , Amateur Observer's Society of New York	June 2022
Toronto, CA	AstroTours Public Science Talk (2) , University of Toronto	Jul. 19, Jun. 22
St. John's, CA	Career Presentation to Grade 5 and 6 Students , St. Mary's Elementary School	Apr. 2022
Virtual	Career Presentation to Grade 7-12 Students , Crescent Collegiate, NL	Feb. 2021

Service, Outreach, and Academic Leadership

CONFERENCE ORGANIZATION

Atlantic Universities Physics and Astronomy Conference

Aug. 2015 - Feb. 2016

Co-CHAIR OF LOC (WITH TYLER DOWNEY)

St. John's, CA

- Co-chair of LOC for regional conference of 100+ undergraduate students across Atlantic Canada
- Responsible for assembling the rest of the LOC and volunteering groups, overseeing conference design and budget, acquiring meeting space and catering, choosing invited speakers, soliciting donations, hosting Women in Science panel, and ensuring the conference ran smoothly.

LEADERSHIP POSITIONS, COMMITTEES, AND SERVICE

Article Referee

2025-

APJ, A&A

Seminar Committee

2025-2026

McWILLIAMS CENTER FOR COSMOLOGY AND ASTROPHYSICS

CMU

- Committee member, responsible for soliciting recommendations for and selecting speakers, & organizing speaker visits

Scholarships and Fellowship Review Panel, Physics & Astronomy NATIONAL SCIENCE AND ENGINEERING RESEARCH COUNCIL OF CANADA • Panel member for evaluating graduate scholarship and postdoctoral fellowship applications. 3 year appointment.	2024,26,27 Canada
McWilliams Fellowship Search Committee MCWILLIAMS CENTER FOR COSMOLOGY AND ASTROPHYSICS • Committee member for evaluating postdoctoral fellowship applications.	2024-2025 CMU
‘Write Together Friends’ Co-organizer MCWILLIAMS CENTER FOR COSMOLOGY AND ASTROPHYSICS • Co-creator and co-organizer of weekly department event to encourage interaction and co-working within the department.	2024-Present CMU
Jamboree Committee Member MCWILLIAMS CENTER FOR COSMOLOGY AND ASTROPHYSICS • Committee member, responsible for organizing the yearly jamboree of flash talks for the department and securing venue for evening event; created standard procedure guide for future committees.	2023 & 2024 CMU
Regular Member, Committee Member GRADUATE ASTRONOMY STUDENT ASSOCIATION • Course & Qualifier Satisfaction Survey Committee (Chair) - Surveying student opinion on courses and qualifier exams and presenting results to faculty • PRISM (LGBTQ+) Committee - Event organization • Social Committee - Event organization	2016-2023 UofT
Regular Member, President (2015-2016) PHYSICS & PHYSICAL OCEANOGRAPHY UNDERGRADUATE SOCIETY • Led the undergraduate society to organize monthly social events, survey students for opinions on course material and program design, and created a scholarship.	2013-2016 MUNL
OUTREACH AND SCIENCE COMMUNICATION	
Scientific Fact Checker BOOK – COSMIC COLLISIONS: SUPERGIANT VS. NEUTRON STAR ISBN-13: 978-1536242256	2024
Media Speaker & Volunteer 2023 TOTAL SOLAR ECLIPSE EVENT, MEMORIAL UNIVERSITY OF NL	2023 MUNL
Volunteer 2017 PARTIAL SOLAR ECLIPSE EVENT & CANADA 150 STAR PARTY, RASC & DUNLAP INSTITUTE	2017 UofT
Volunteer MATH DEPARTMENT GIRLS IN STEM EVENT	2017 UofT
Volunteer ASTROTOURS & ASTRO ON TAP	2016-2020 UofT

Media, Interviews, & Press

Blog Post	Searching for Betelgeuse's Buddy with Chandra , Anna O'Grady & Brendan O'Connor	Nov. 2025
Press Rel.	X-Ray Study Reveals New Details About Betelgeuse's Elusive Companion Star , Amy Pavlak Laird, CMU Mellon College of Science	Oct. 2025
Interview	Anna O'Grady McWilliams Postdoctoral Fellow , Gale Force Wins Podcast, Ep 225	Feb. 2025
Interview	Holiday Question Show , Quirks and Quarks, CBC	Dec. 2023
Video	Stellar Struggles: Frank Kameny's fight for LGBTQ+ equality , Cosmos From Your Couch	June 2023
Docu.	Astronomer Searches for the Universe's Largest Stars , Imagine This	June 2023
Article	NL PhD student makes explosive discovery amongst the stars , Peter Jackson, Saltwire	June 2021
Article	Stellar alumna: Astronomy scholar discovers new type of star , Kelly Foss, MUN Gazette	June 2021
Article	On the Cusp of a New Understanding of the Stars , Emily Deibert, Research2Reality	Apr. 2021
Press Rel.	UofT Student Discovers Rare Star Population , Meaghan MacSween, Dunlap Institute	Mar. 2021

Collaborations and Professional Organizations

BLOeM BINARITY AT LOW METALLICITY COLLABORATION	2024-Present
YSE YOUNG SUPERNOVA EXPERIMENT COLLABORATION	2021-2023
IAU INTERNATIONAL ASTRONOMY UNION	2024-Present
CASCA CANADIAN ASTRONOMICAL SOCIETY SOCIÉTÉ CANADIENNE D'ASTRONOMIE	2019-Present
AAS AMERICAN ASTRONOMICAL SOCIETY	2022-Present

Teaching Experience

As a postdoc, I have instructed 50+ undergraduate students as a **guest lecturer** for 4 lectures, one of which I **designed myself**, and I attended an APS/AAS teaching workshop. As a graduate student, I took a course on the pedagogy of teaching, and was a teaching assistant (TA) for **16 courses**. From 2018-2023, I was the **Head Marker** for two of the **largest** undergraduate courses in Canada (~ **1500 students** per semester). In this role I was responsible for organizing the duties of a team of 20+ TAs and overseeing the grading of all students. I also **designed and delivered** a lecture to summer research students. Finally, I was a TA for a graduate course in 2022 and a undergraduate lab course during my senior undergraduate year.

CMU	PHYS 33120: Science and Science Fiction , Guest Lecturer (4 Lectures)	2025
UofT	Summer Undergraduate Research Program , Guest Lecturer (1 lecture)	2022
UofT	AST 301: Observational Astronomy , Graduate TA (x4)	2019-2022
UofT	AST 1410: Stars (Graduate Course) , Senior Graduate TA, Half semester appointment	2021
UofT	AST 101/201: The Sun & Its Neighbours/Stars & Galaxies , Head Marker (x5.5), Grad. TA (x6)	2016-2023
MUNL	PHYS 1020: Introductory Physics I , Undergraduate TA	2015

Publication List

21 total publications; 5 first-author, 1 invited review, 1 white paper. 60+ citations from significant contribution articles; over 400 citations total. Underlined authors indicate student supervision.

INVITED REVIEWS (1)

1. **O’Grady, A.J.G.**; Moriya, T.J.; Renzo, M.; Vigna-Gomez, A.; “Thorne-Zytkow Objects”, 2024, *Reference Module in Materials Science and Materials Engineering*, Elsevier, published as a chapter in Elsevier’s *Encyclopedia of Astrophysics* (2026), edited by I. Mandel, 17pp

SIGNIFICANT CONTRIBUTION (8)

8. Goldberg, J.; Joyce, M.; **O’Grady, A.J.G.**; et al., “Disco in the Dust: Reflected light at the bow shock around Betelgeuse’s companion explains its observed luminosity”, 2026, Submitted to *AAS Journals*, arXiv:2608.11672, 17pp
7. Martinez, M.; **O’Grady, A.J.G.**; Breivik, K.; Chen, G., “Properties of Core Collapse Supernovae from Binary Population Synthesis”, 2025, Submitted to *ApJ*, arXiv:2511.23285, 30pp
6. **O’Grady, A.J.G.**, “The Kinematic Properties of TZO Candidate HV 11417 with Gaia DR3”, 2025, Published in *The Open Journal of Astrophysics*, 10.33232/001c.155625, 5pp
5. **O’Grady, A.J.G.**, O’Connor, B.; Goldberg, J.; et al., “Betelgeuse’s Buddy: X-Ray Constraints on the Nature of α Ori B”, 2025, *ApJ*, 992, 107, 12pp
Press Release/Media: [Carnegie Mellon University](#)
4. Goldberg, J.; **O’Grady, A.J.G.**; et al., “Betelgeuse, Betelgeuse, Betelgeuse, Betel-buddy? Constraints on the dynamical companion to α Orionis from HST”, 2025, *ApJ* 994, 101, 15 pp
3. **O’Grady, A.J.G.**; Drout, M.R.; Neugent, K.F.; et al., “Binary Yellow Supergiants in the Magellanic Clouds. I: Photometric Candidate Identification”, 2024, *ApJ*, 975, 29, 21pp
2. **O’Grady, A.J.G.**; Drout, M.R.; Gaensler, B.M.; et al., “Cool, Luminous, and Highly Variable Stars in the Magellanic Clouds. II: Spectroscopic and Environmental Analysis of Thorne-Zytkow Object and super-AGB Star Candidates”, 2023, *ApJ*, 943, 18, 26pp
1. **O’Grady, A.J.G.**; Drout, M.R.; Shappee, B.J.; et al., “Cool, Luminous, and Highly Variable Stars in the Magellanic Clouds from ASAS-SN: Implications for Thorne-Zytkow Objects and super-Asymptotic Giant Branch Stars”, 2020, *ApJ*, 901, 2, 32pp

OTHER CONTRIBUTIONS (12)

12. Zhang, R.; Breivik, K.; **O’Grady, A.J.G.**, “Siwarha as a Neutron Star: The Kinematics of a Red Supergiant-Neutron Star Binary System”, 2026, *RNAAS*, 10, 8
11. Gilkis, A.; et al. (5 co-authors, including **O’Grady, A.J.G.**), “Hubble as a Unique Discovery Engine of the Fate of Massive Stars and Black Hole Formation”, 2026, *White Paper for ‘Building a Roadmap for Hubble science into the 2030s’*, arXiv:2606.10710
10. Gilkis, A.; et al. (5 co-authors, including **O’Grady, A.J.G.**), “Hot blue progenitors of stellar-mass black holes”, 2026, Submitted to *MNRAS*, arXiv:2604.12868
9. Dorn-Wallenstein, T.Z.; et al. (7 co-authors, including **O’Grady, A.J.G.**), “A Spectroscopic Hunt for Post-Red Supergiants in the Large Magellanic Cloud II: Turbulent Line Broadening in the Spectra of LMC Yellow Supergiants”, 2025, *ApJ*, 991, 173
8. Patrick, L.R.; et al. (39 co-authors including **O’Grady, A.J.G.**), “Binarity at Low Metallicity (BLOeM): The multiplicity properties and evolution of BAF-type supergiants”, 2025, *A&A* 698, A39
7. Shenar, T.; et al. (77 co-authors including **O’Grady, A.J.G.**), “Binarity at Low Metallicity (BLOeM): A spectroscopic VLT monitoring survey of massive stars in the SMC”, 2024, *A&A*, 690, A289
6. Drout, M.R.; et al. (7 co-authors including **O’Grady, A.J.G.**), “An observed population of intermediate-mass helium stars that have been stripped in binaries”, 2023, *Science*, 382, 6676
Press Release/Media: [University of Toronto](#), [Carnegie Observatories](#)
5. Kilpatrick, C.D.; et al. (25 co-authors including **O’Grady, A.J.G.**), “Type II-P supernova progenitor star initial masses and SN 2020jfo: direct detection, light-curve properties, nebular spectroscopy, and local environment”, 2023, *MNRAS*, 524, 2
4. Aleo, P.D.; et al. (87 co-authors including **O’Grady, A.J.G.**), “The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae”, 2023, *ApJS*, 226, 1

3. Chan, H-S.; et al. (6 co-authors including **O’Grady, A.J.G.**), “[Searching for Anomalies in the ZTF Catalog of Periodic Variable Stars](#)”, 2022, *ApJ*, 932, 2
2. Kessler, T.; et al. (30 co-authors including **O’Grady, A.J.G.**), “[Models and Simulations for the Photometric LSST Astronomical Time Series Classification Challenge \(PLAsTiCC\)](#)”, 2019, *PASP*, 131, 1003
1. Booth, I.; Kunduri, H.K.; **O’Grady, A.J.G.**, “[Unstable marginally outer trapped surfaces in static spherically symmetric spacetimes](#)”, 2017, *Physical Review D*, 96, 2